

Quenzafil Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-231/C

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Cleaning verification swab results were below the health-based exposure limits derived for the product. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process parameters identified during development.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

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Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

3.2.P.4 Excipient Control Strategy

For the commercial formulation F-231/C, Ligamed MF-2-V (Magnesium stearate, lubricant) is used.

Grade Selection Rationale

The formulation development rationale for Ligamed MF-2-V rests on tighter particle size distribution supporting content uniformity.

The approach described herein is consistent with IPEC Excipient Qualification Guide.

Current Commercial Specification

Component	Function	Grade per current specification
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Magnesium stearate	lubricant	Kosher Passover grade HyQual
Reference		SPEC-EX-8355 Rev 08

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.